

RayGlides 3W/4W EV Kit Cooling Controller - Requirements and Assumptions

1. Project Objective

Design a production-grade active cooling controller board for the electronics kit enclosure of RayGlides 3-wheeler and 4-wheeler electric cargo vehicles. The system reads temperature data and automatically adjusts a 12V brushless DC cooling fan using PWM, maintaining system reliability under high ambient and EMI conditions.

2. Electrical Specifications

- * **Nominal Vehicle Input Voltage**: 48V DC
- * **Operating Input Voltage Range**: 36V to 72V DC (handles battery charge/discharge cycles)
- * **Transient Protection**: Clamping spikes up to 100V (e.g. from motor regenerative braking/switching surges)
- * **Reverse Polarity Protection**: Series Schottky barrier diode or P-channel MOSFET (selected: 100V Schottky diode for simplicity and low leakage)
- * **Overcurrent Protection**: 2A slow-blow fuse on the input stage
- * **Low-Voltage Rails**:
 - * **12V DC**: For powering the cooling fan (peak 600mA buck converter)
 - * **5V DC**: For active sensors and secondary peripherals (2A synchronous buck)
 - * **3.3V DC**: Clean, low-noise LDO output for ESP32-S3 and analog reference (600mA LDO)

3. Microcontroller & Sensor Interfaces

- * **MCU Target**: ESP32-S3-N16R8
- * **Control GPIO (Fan PWM)**: GPIO 18 (LEDC PWM channel 1, 5 kHz frequency)
- * **Sensor Input Pin**: GPIO 4 (ADC1_CH3, analog voltage reading)
- * **Temperature Sensor**: LM35DT analog temperature sensor (active, 10mV/ deg C linear response, TO-220 package for direct heatsink mounting)
- * **Measurement Range**: 0 deg C to 100 deg C (corresponding output 0V to 1.0V)
- * **Programming Port**: Native USB-C direct flashing interface via GPIO 19 and 20.
- * **Communication Bus**: Integrated TJA1050 CAN bus transceiver via GPIO 9 and 10.
- * **ADC Calibration Reference**: Onboard LM4040AIM3-1.0 shunt voltage reference for ESP32-S3 ADC calibration.
- * **Memory Isolation**: Complete isolation of the GPIO 26-37 range used by the Octal PSRAM.

4. Environmental and Mechanical Design

- * **Operating Ambient Temperature**: -20 deg C to +65 deg C (automotive cabin/enclosure ambient)
- * **Enclosure Rating**: IP65-equivalent kit box with baffled cooling fan intake/exhaust vents
- * **Vibration Resistance**: Production locking connectors (Molex Micro-Fit 3.0 or equivalent) to prevent disconnection during cargo operations
- * **Board Dimensions**: 80mm x 55mm double-sided FR4 PCB (1.6mm thickness, 2 oz/ft² copper)

5. Control Philosophy

- * **Configurable Thresholds**:
- * **Fan Turn-On Temperature**: 45 deg C
- * **Fan Turn-Off Temperature**: 40 deg C (5 deg C hysteresis to prevent rapid ON/OFF cycling)
- * **Fan Scaling**: Dynamic scaling from 30% duty cycle at the ON threshold to 100% duty cycle at 55 deg C (linear scaling implemented in firmware).
- * **Emergency Thermal Shutdown**: Initiated if temperature exceeds 65 deg C.